

Supplementary Material for SAW-Bench-CS: A Benchmark for SpotBugs Warning Context Selection

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Note. This supplementary material is provided for double-anonymous review and is not part of the 20-page main submission.

A Related-Work Gap Map

The gap map supports the Related Work section by making the comparison dimensions explicit. It is placed here rather than in the main paper so that the main text can summarize the benchmark gap in prose.

B Dataset Detail Tables

This section gives the dataset details that support the main-paper summary tables and figures. The tables report snippet types, warning categories, the difficulty-tier rule, and per-project counts without changing the released JSONL schema.

Table 1 Candidate snippet types and essential-label rates used to interpret evidence coverage.

Snippet Type	Count	% Essential
warning_line	403	100.0%
enclosing_method	403	100.0%
enclosing_class	403	12.9%
caller	640	1.4%
callee	472	10.8%
field_or_type_declaration	340	62.9%
annotation_or_contract	69	14.5%
test	225	0.0%
similar_code	144	0.0%

Table 2 Warning-category distribution and average candidate-set size in the released benchmark.

Category	Warnings	Avg. Snippets
Bad practice	139	7.7
Security	123	8.0
Multithreaded correctness	39	7.7
Performance	36	7.6
Correctness	25	6.8
Nullness	16	6.9
Dodgy code	11	7.4
Equality/hash-code	11	8.2
Resource management	3	6.0



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Benchmark family	Real projects	Warning labels	Candidate evidence	Evidence labels	Artifact scripts	Context cost
Bug benchmarks	Y	N	P	N	Y	N
Vulnerability data	Y	Y	P	N	P	N
Warning filtering	Y	Y	N	N	P	N
Repository LLM tasks	Y	N	P	N	Y	P
RAG/context papers	P	N	P	N	P	P
SAW-Bench-CS	Y	Y	Y	Y	Y	Y

Y=present P=partial/implicit N=absent

■ **Figure 1** Benchmark gap map comparing SAW-Bench-CS with related benchmark families.

■ **Table 3** Recall@3 per SpotBugs category on the held-out test split. LF = Local-first; TP = Type-priority; Emb = all-MiniLM-L6-v2. Categories with $n \leq 3$ should be interpreted cautiously.

Category	n	BM25	Emb	LF	TP
Bad practice	15	0.378	0.389	0.756	0.911
Correctness	6	0.556	0.611	0.778	0.944
Dodgy code	3	0.500	0.444	0.889	0.889
Equality/hash-code	8	0.333	0.375	0.708	0.875
Multithreaded correctness	12	0.486	0.542	0.889	0.972
Nullness	7	0.571	0.667	0.857	0.810
Performance	13	0.603	0.449	0.795	0.846
Resource management	1	0.333	0.667	0.667	0.667
Security	12	0.486	0.514	0.750	0.889
All	77	0.481	0.489	0.792	0.892

16 The rule is exhaustive and mutually exclusive on the labeled benchmark. Tier priority is
 17 repository-aware > cross-reference > local. If a future warning has no essential snippets, the
 18 rule falls back to the helpful set and applies the same priority order.

■ **Algorithm 1** Difficulty-tier assignment used to summarize whether essential evidence is local, cross-reference, or repository-aware.

Input: Warning w with essential snippet set
 $E_w = \{s \in \text{candidates}(w) : \text{label}(s) = \text{ESSENTIAL}\}.$

Output: Difficulty tier $t_w \in \{\text{LOCAL}, \text{CROSS-REFERENCE}, \text{REPOSITORY-AWARE}\}.$

```

1  $T_E \leftarrow \{\text{type}(s) : s \in E_w\}$ 
2  $L \leftarrow \{\text{warning\_line}, \text{enclosing\_method}\}$ 
3  $C_1 \leftarrow \{\text{enclosing\_class}, \text{caller}, \text{callee}\}$ 
4  $C_2 \leftarrow \{\text{field\_or\_type\_declaration}, \text{annotation\_or\_contract}\}$ 
5  $C \leftarrow C_1 \cup C_2$ 
6  $R \leftarrow \{\text{test}, \text{configuration}, \text{similar\_code}, \text{history\_or\_diff}\}$ 
7 if  $T_E \cap R \neq \emptyset$  then
8   | return REPOSITORY-AWARE
9 end
10 if  $T_E \cap C \neq \emptyset$  then
11   | return CROSS-REFERENCE
12 end
13 return LOCAL

```

■ **Table 4** Per-project warning counts underlying the project-based train/validation/test split.

Split	Project	Warnings
Train	commons-scxml	52
	commons-configuration	38
	commons-lang	36
	commons-beanutils	33
	commons-net	28
	commons-imaging	18
	commons-dbutils	11
	commons-cli	10
	commons-exec	8
	commons-codec	7
	commons-crypto	5
	commons-daemon	2
	commons-csv	1
Validation	mybatis-3	16
	junit4	13
	jackson-core	12
	commons-io	10
	commons-ognl	8
	commons-dbcp	5
	commons-validator	4
	HikariCP	4
	commons-pool	2
	maven-shade-plugin	2
	commons-text	1
Test	commons-compress	20
	lettuce-core	17
	jackson-databind	13
	commons-bcel	8
	commons-collections	7
	commons-jexl	7
	jsoup	3
	jackson-annotations	1
	mybatis-spring	1
Total		403

C Baseline and Protocol Detail Tables

This section records secondary protocol information for reproducing or extending the baseline experiments. The main paper reports only the core held-out results; the tables below give cost, diagnostic, type-frequency, and token-budget details.

Table 5 Local cost budget for benchmark construction and default baseline evaluation.

Stage	Local wall-clock cost	Hardware
Candidate generation (per warning, p95)	11 s	16-core CPU
Candidate generation (full benchmark, 403 warnings)	recorded in artifact log	16-core CPU
SpotBugs analysis (full benchmark)	recorded in artifact log	16-core CPU
BM25 ranking (test split, 77 warnings)	<1 s	single CPU
Embedding ranking (test split, 77 warnings, all-MiniLM-L6-v2 local cache)	recorded in artifact log	local CPU/GPU; no endpoint
Type-priority ranking (test split, 77 warnings)	1 s	single CPU
LLM reranking	endpoint-dependent	configured LLM endpoint
End-to-end test-split evaluation (default baselines)	<1 s	single CPU

Table 6 Additional diagnostic metrics for analyzing context-selection behavior beyond headline Recall@ k .

Metric	Definition	Recommended Plot
Evidence coverage@ k	Useful snippets in top- k / all useful snippets	Line curve over k
Token-normalized utility	Graded relevance gain per 1K selected tokens	Utility scatter
Distractor rate@3	Irrelevant snippets in top-3 / 3	Stacked top-3 composition bars
Non-local recovery@3	Non-local essential snippets in top-3 / all non-local essential snippets	Type heatmap

The no-local ablation confirms that local evidence explains a large fraction of the headline score. Removing `warning_line` and `enclosing_method` candidates reduces Recall@3 for all methods, including type-priority (0.892 to 0.313) and Local-first (0.792 to 0.198). Type-priority nevertheless remains the strongest method under this restricted setting, suggesting that the benchmark still distinguishes rankers after local evidence is removed.

The hard-negative stress test adds injected irrelevant snippets to the candidate pool and evaluates recall against the original essential labels. Local-first is unaffected at Recall@3 and selects no hard negatives in the top three, while embedding and type-priority degrade moderately. BM25 is more sensitive, dropping from 0.481 to 0.325 Recall@3 and selecting injected hard negatives in 22.9% of top-3 positions. These results reduce the concern that all headline rankings collapse under additional distractors, but they also show that hard-negative robustness varies by method.

The enriched query ablation shows that adding warning-line text substantially improves BM25 and MiniLM over the default headline query, but both remain well below type-priority and the learned LTR baseline.

Candidate-generation coverage.

On the held-out test split, all 218 released essential snippets are present in the candidate pools, including all 64 non-local essentials. The upstream miss rate with respect to released labels is therefore 0.0%. Coverage is 100% for warning lines (77/77), enclosing methods (77/77),

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■ **Table 7** Essential context-type frequency used to interpret the strength of type-priority ranking.

Snippet Type	Warnings with Essential Snippet
warning_line	100.0%
enclosing_method	100.0%
field_or_type_declaration	53.1%
enclosing_class	12.9%
callee	12.7%
annotation_or_contract	2.5%
caller	2.2%
test	0.0%
history_or_diff	0.0%
configuration	0.0%
similar_code	0.0%

■ **Table 8** Prompt token budget by context strategy for the held-out RQ4 protocol.

Context Strategy	Avg. Tokens	Snippets/warning	Tokens/snippet	Reduction vs. Full
Full candidate context	2,124	7.8	273	–
Local-only context	1,494	2.0	747	30%
Selected context (Type-priority top-3)	1,507	3.0	502	29%

42 field/type declarations (36/36), enclosing classes (16/16), callees (10/10), annotation/contract
 43 snippets (1/1), and callers (1/1). This analysis is bounded by the released labels and does
 44 not rule out unlabeled evidence outside the candidate pools.

45 **Type-priority failure analysis.**

46 Type-priority fails to recover all top-3 essential evidence for 25/77 test warnings (32.5%).
 47 All failures have Recall@3=0.667, meaning that two of three essential snippets are re-
 48 covered. Every missed snippet is non-local: 15 are `enclosing_class`, 8 are `callee`, 1 is
 49 `annotation_or_contract`, and 1 is `caller`. This suggests that rule-family-specific priors
 50 could improve over global type priors, especially for class-level and callee evidence.

■ **Table 9** Leave-one-project-out robustness summary across held-out projects.

Method	Mean R@3	SD	Min	Max	Worst project
BM25	0.431	0.144	0.000	0.667	commons-text
Embedding (all-MiniLM-L6-v2)	0.478	0.164	0.000	1.000	commons-text
Local-first	0.780	0.091	0.667	1.000	HikariCP
Type-priority	0.910	0.076	0.667	1.000	mybatis-spring

■ **Table 10** Bootstrap 95% confidence intervals for held-out context-selection metrics.

Method	R@3	nDCG@5	Non-local R@3
Local-first	0.792 [0.758, 0.827]	0.961 [0.952, 0.971]	0.250 [0.148, 0.361]
BM25	0.481 [0.424, 0.537]	0.833 [0.811, 0.855]	0.359 [0.246, 0.478]
Embedding (all-MiniLM-L6-v2)	0.489 [0.433, 0.543]	0.857 [0.837, 0.878]	0.375 [0.258, 0.493]
Type-priority	0.892 [0.857, 0.926]	0.990 [0.985, 0.995]	0.609 [0.485, 0.727]

■ **Table 11** Merge-rule sensitivity across representative rankers on the held-out split. Values are Recall@3.

Merge rule	#Essential	BM25	Local-first	Type-priority
Current merge	218	0.481	0.792	0.892
A only	159	0.511	0.978	1.000
B only	192	0.481	0.721	0.866
Strict agreement essential	126	0.528	0.972	1.000
Lower-bound disagreement	126	0.528	0.972	1.000
Useful binary	567	0.425	0.425	0.428

■ **Table 12** No-local ablation on the held-out split. The no-local setting removes `warning_line` and `enclosing_method` candidates and evaluates the remaining warnings with non-local essential evidence.

Method	Full R@3	No-local R@3	No-local nDCG@5	No-local non-local R@3	R@3 change
Random	0.416	0.208	0.639	0.625	-0.207
Local-first	0.792	0.198	0.649	0.594	-0.594
BM25	0.481	0.177	0.630	0.531	-0.303
Embedding (all-MiniLM-L6-v2)	0.489	0.219	0.647	0.656	-0.270
Type-priority	0.892	0.313	0.701	0.938	-0.579

■ **Table 13** Injected hard-negative stress test on the held-out split. Essential recall is computed against the original essential labels; hard-negative rate reports the fraction of injected hard negatives selected in the top- k .

Method	Clean R@3	Stress R@3	Δ R@3	nDCG@5	Hard-neg@3	Hard-neg@5
Random	0.416	0.216	-0.199	0.502	0.342	0.356
Local-first	0.792	0.792	0.000	0.919	0.000	0.086
BM25	0.481	0.325	-0.156	0.650	0.229	0.231
Embedding (all-MiniLM-L6-v2)	0.489	0.455	-0.035	0.777	0.069	0.117
Type-priority	0.892	0.848	-0.043	0.888	0.117	0.195

■ **Table 14** Query-formulation ablation for BM25 and all-MiniLM-L6-v2 embedding retrieval on the held-out split. Values are Recall@3 and nDCG@5.

Query variant	BM25 R@3	BM25 nDCG@5	Emb R@3	Emb nDCG@5
Message + rule + category	0.481	0.833	0.489	0.857
Message + rule + category + warning line	0.595	0.875	0.591	0.883

■ **Table 15** Hybrid and learned ranking diagnostics on the held-out split.

Method	Recall@3	nDCG@5	Non-local R@3
LTR (pairwise logistic)	0.905	0.993	0.656
TP+BM25 rerank	0.896	0.989	0.625
Type-priority	0.892	0.990	0.609
Local-first	0.792	0.961	0.250
RRF(BM25+MiniLM+TP)	0.671	0.920	0.313
Convex BM25/MiniLM, $\alpha = 0.3$	0.507	0.852	0.391
RRF(BM25+MiniLM)	0.487	0.848	0.359
MiniLM	0.489	0.857	0.375
BM25	0.481	0.833	0.359

■ **Table 16** Largest pairwise logistic LTR coefficients. Positive values increase ranking score.

Feature	Coefficient
is_local	2.907
test	-2.538
enclosing_method	2.238
caller	-2.187
warning_line	1.873
log_tokens	-1.682
similar_code	-1.272
bm25_score	0.890
field_or_type_declaration	0.617
emb_cosine	0.239

■ **Table 17** LTR feature ablation on the held-out split. R@3 is Recall@3 and NLR@3 is Non-local Recall@3.

Ablation	R@3	NLR@3
Full 16-feature LTR	0.905	0.656
No type one-hot	0.801	0.281
No text features	0.905	0.656
Type only	0.905	0.656
Text only (BM25+Emb)	0.509	0.484
Is-local only	0.792	0.250

■ **Table 18** LTR regularization sensitivity. The deployed setting uses $C = 1.0$, selected a priori as the sklearn default.

C	Val R@3	Val R@5	Test R@3	Test R@5
0.01	0.922	0.974	0.905	0.991
0.10	0.922	0.983	0.905	0.987
1.00	0.922	0.983	0.905	0.991
10.0	0.922	0.983	0.909	0.991
100.0	0.922	0.983	0.909	0.991

■ **Table 19** Hard-negative sensitivity as injected negatives increase relative to the original candidate-pool size. Values are Recall@3.

Injection ratio	Local-first	BM25	MiniLM	Type-priority
0.00	0.792	0.481	0.489	0.892
0.25	0.792	0.418	0.470	0.853
0.50	0.792	0.362	0.463	0.844
1.00	0.792	0.325	0.455	0.844

D Detailed Annotation Statistics

Table 20 presents the raw agreement counts between the two A/B artifact-labeling passes across the 3,099 candidate snippets. Most disagreements occur between *essential* and *helpful*, which is the main boundary used to interpret annotation uncertainty in the paper.

■ **Table 20** Released A/B context-relevance agreement counts for all 3,099 candidate snippets.

Pass A	Pass B			Total
	<i>essential</i>	<i>helpful</i>	<i>irrelevant</i>	
<i>essential</i>	690	199	0	889
<i>helpful</i>	278	1,230	477	1,985
<i>irrelevant</i>	0	214	11	225
Total	968	1,643	488	3,099

Table 21 reports the overall Cohen’s κ and the prevalence-adjusted bias-adjusted kappa (PABAK) recomputed from `annotation/annotator_passes.jsonl`. PABAK uses the multi-class generalization $(n \cdot p_o - 1)/(n - 1)$ [1], where $n = 3$ is the number of relevance classes and $p_o = 0.6231$ is the observed agreement. With this formula, PABAK is 0.435.

■ **Table 21** Agreement statistics recomputed from the released A/B annotation-pass records.

Metric	Value
Overall Cohen’s κ	0.326
Prevalence-Adjusted κ (PABAK, 3-class)	0.435

E Explanation Quality Rubric (RQ4)

This appendix defines the rubric used by the RQ4 protocol in the main paper. Table 22 gives anchors for every 1–5 score level across the four criteria, including explicit anchors for intermediate scores. The artifact applies the rubric with two deterministic scorers in `scripts/generate_rq4_artifacts.py` and ships the long-format scores in `results_reproduced/explanations_scored.csv`. On the full 231-explanation set, deterministic scorer agreement is Pearson $r = 0.601$ for per-explanation quality and quadratic-weighted $\kappa = 0.571$ for rounded four-criterion means. These statistics characterize the reproducible scorer used for protocol verification; they are not human inter-rater reliability.

Future RQ4 studies should use criterion 3 (factuality) only for claims about program behavior, not for judgments about whether the warning is a true defect. This keeps context selection separate from warning validity, following the construct distinction in the main paper. Such studies should release generated explanations, prompt records, per-rater scores, and calibration logs alongside aggregate tables.

■ **Table 22** Full 1–5 RQ4 explanation-quality rubric used by the deterministic protocol scorers.

Criterion	Score	Anchor Description
Cause Identification	5	Explicitly names the specific variable, condition, or API contract that SpotBugs flagged.
	4	Names the right code element but understates or misorders the conditions that trigger the warning.
	3	Mentions the general area of the bug but misses the specific root cause.
	2	Identifies a related symptom or a different defect on the same line.
	1	Incorrectly identifies the cause or is generic (e.g., “there is a bug”).
Evidence Usage	5	Correctly references specific lines or symbols from the provided context snippets.
	4	References relevant symbols and connects them to the warning, with one minor mis-attribution.
	3	Mentions relevant symbols but does not connect them to the warning logic.
	2	References evidence that is in the context but not relevant to the warning.
	1	No reference to evidence; purely hallucinates behavior.
Factuality	5	All claims about program behavior are supported by the provided context.
	4	One minor unsupported assumption that does not invalidate the explanation.
	3	Contains a small number of unsupported claims that weaken but do not break the explanation.
	2	Contains an unsupported claim that the reader would have to verify before acting.
	1	Contains major hallucinations or claims that contradict the source code.
Conciseness	5	Single paragraph, no filler text, directly answers the cause.
	4	One paragraph plus a brief restatement of the warning message.
	3	Includes some redundant background or repeats the warning message.
	2	Multiple paragraphs of background before reaching the cause.
	1	Extremely verbose or contains large amounts of irrelevant boilerplate.

73 **F** LLM Reranker Prompt Template

74 The optional endpoint LLM reranker described in the main paper uses the fixed JSON-
 75 ordering prompt below. The runner fills in inputs without per-warning customization,
 76 and the model output is parsed as a JSON object containing an `order` list. The default
 77 endpoint model identifier in the code is `phi-3-mini-instruct`, but any OpenAI-compatible
 78 or Ollama-compatible endpoint can be used.

79 You are scoring candidate code snippets for usefulness in explaining a static
 80 analysis warning. Return a JSON list of 'snippet_id' strings ordered most
 81 useful first. Use only the snippets provided.

82
 83 Warning ({rule_id}): {message}

84 Warning line: {line_text}

85
 86 Candidates:

87 {candidates}

88
 89 Return exactly: {"order": ["s01", ...]}

90 Decoding and model behavior are endpoint-dependent. The reranker is applied to the
 91 top five BM25 candidates per warning and is skipped by default when `SAW_LLM_ENDPOINT` is
 92 not configured.

93 — References —

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